

asset classes, as does value.³³ It can be part of a long-run investor's opportunistic strategy (the Merton (1969) long-run hedging demand portfolio).

Momentum is often used as an investment factor, added onto the Fama–French model.³⁴

$$E(r_i) = r_f + \beta_{i,MKT}E(r_m - r_f) + \beta_{i,SMB}E(SMB) + \beta_{i,HML}E(HML) + \beta_{i,WML}E(WML) \quad (7.4)$$

The same intuition applies as with the Fama–French model. The momentum beta, $\beta_{i,WML}$, is centered around zero. Winner stocks have positive momentum betas; their risk premiums are adjusted upward using equation (7.4). Loser stocks have negative momentum betas; their risk premiums are adjusted downward. The market, neither a relative winner nor a relative loser, is simply the market.

Characterizing Momentum Risk

Figure 7.7 shows that despite the large return, on average, of momentum strategies, momentum is prone to periodic crashes. Some of these have lasted for extended periods. Daniel and Moskowitz (2012) examine these in detail. Of the eleven largest momentum crashes, seven occurred during the Great Depression in the 1930s, one occurred in 2001, and the other three occurred during the financial crisis in 2008. The loser stocks then were tanking financials: Citi, Bank of America, Goldman, and Morgan Stanley, and some others hard hit by circumstances, like General Motors. Loser stocks have a tendency to keep losing, and lose they would have, were it not for Uncle Sam riding to their rescue. Government bailouts put a floor underneath the prices of these stocks, and they consequently skyrocketed. Since momentum strategies were short these stocks, momentum investors experienced large losses when these stocks rebounded. It is notable that the other big momentum drawdowns were concentrated during the Great Depression when policymakers also had great influence on asset prices. Momentum seems to reflect monetary policy and government risk during extraordinary times. These have also been times of high volatility.

What else explains momentum? Tantalizing suggestions in the literature suggest that at least some portion of momentum profits correlates with macro factors. Momentum profits, for example, vary over the business cycle and depend on the state of the stock market, and there is a link with liquidity.³⁵ In the rational story of momentum (which is still far from being fully fleshed out in the literature), asset owners should examine how they behave facing the various sources of macro risk discussed earlier.

³³ See Blitz and Van Vliet (2008).

³⁴ Carhart (1997) was the first to do this.

³⁵ See Chordia and Shivakumar (2002), Cooper, Gutierrez, and Hameed (2004), and Pástor and Stambaugh (2003), respectively.